

FUNDAMENTALS OF MECHANICAL ENGINEERING

- Mechanical engineering is a branch of engineering that deals with the design, construction, and operation of machines and mechanical systems.
- It is the application of physical principles for analysis, design, manufacturing, and maintenance of mechanical systems.
- The scope of mechanical engineering includes the design, analysis, and development of machines, tools, and processes.
- Mechanical engineers use a variety of materials, such as metals, plastics, composites, and ceramics, to design and manufacture products.
- Mechanical engineers must understand the principles of physics and mathematics to be able to design and analyze components and systems.
- Mechanical engineers must be able to use computer-aided design (CAD) and computer-aided manufacturing (CAM) software to design and manufacture products.
- Mechanical engineers must understand the principles of thermodynamics, fluid mechanics, and heat transfer to be able to design and analyze components and systems.
- Mechanical engineers must be able to use finite element analysis (FEA) to analyze the stress and strain on components and systems.
- Mechanical engineers must be able to use computer-aided engineering (CAE) tools to analyze and optimize components and systems.
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Unit 1 - Introduction to Mechanics

1. Mechanics is the branch of physical science that studies the motion of objects and the forces that cause them to move.
2. It is a fundamental part of physics and is closely related to other physical sciences such as chemistry, astronomy, and geology.
3. Mechanics is divided into two main branches: classical mechanics and quantum mechanics. Classical mechanics deals with the motion of macroscopic objects while quantum mechanics deals with the motion of subatomic particles.
4. The fundamental principles of mechanics are Newton's laws of motion and the law of conservation of energy.
5. Newton's laws of motion describe how objects move and interact with each other, while the law of conservation of energy states that energy can neither be created nor destroyed.
6. These laws are used to describe and predict the motion of objects in a variety of situations, from the motion of planets to the motion of a car.
7. Mechanics is used in many fields, including engineering, aerospace, robotics, and medicine.

Force Moment and Couple

- Force moment is the product of a force and the distance from the point of application of the force to the point at which the moment is considered.
- Couple is a system of two parallel forces of equal magnitude and opposite direction.
- The moment of a couple is equal to the product of one of the forces and the perpendicular distance between them.
- The moment of a force system is the sum of the moments of the individual forces.
- The moment of a force about a point is equal to the sum of the moments of the individual components of the force about the same point.
- The moment of a force about an axis is equal to the sum of the moments of the individual components of the force about the same axis.
- The moment of a force about an axis is equal to the product of the force and the perpendicular distance from the axis to the point of application of the force.
- The moment of a couple about an axis is equal to the product of one of the forces and the perpendicular distance from the axis to the line of action of the couple.

Principle of Transmissibility

1. Transmissibility is a mechanical engineering principle that states that the forces and moments acting on a system are equal to the forces and moments transmitted to the system.
2. This principle is used to analyze the behavior of machines and systems under load, and to determine the forces and moments that the components of a system must withstand.
3. Transmissibility can be used to calculate the forces and moments that are transmitted through a system when a given load is applied.
4. It is also used to determine the effects of friction, inertia, and other forces on a system.
5. Transmissibility is an important concept in the study of mechanics, and it is essential for understanding the behavior of machines and systems under load.

Varignon's Theorem

Varignon's theorem is a fundamental law of mechanics that states that the resultant of a system of forces is equal to the vector sum of the individual forces. This theorem is applicable to both two-dimensional and three-dimensional systems of forces.

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- The theorem is applicable to both two-dimensional and three-dimensional systems of forces.
- Varignon's theorem can be used to calculate the resultant force and its direction when the magnitude and direction of each force in the system is known.
- The theorem can also be used to calculate the magnitude and direction of each force in the system when the magnitude and direction of the resultant force is known.
- Varignon's theorem can be used to calculate the moment of a force about a point when the magnitude and direction of the force is known.

Resultant of Force System - Concurrent and Non-Concurrent Coplanar Forces

1. A force system is a combination of two or more forces acting on a body.
2. The resultant of a force system is the single force that has the same effect as the combination of the forces.
3. When the forces in a system are concurrent, they all pass through the same point.
4. The resultant of a concurrent force system can be determined by the parallelogram law of forces or the triangle law of forces.

5. The parallelogram law states that the resultant of two concurrent forces is equal to the diagonal of the parallelogram formed by the two forces.
6. The triangle law states that the resultant of two concurrent forces is equal to the sum of the two forces and the angle between them.
7. When the forces in a system are non-concurrent, they do not pass through the same point.
8. The resultant of a non-concurrent force system can be determined by the polygon law of forces.
9. The polygon law states that the resultant of two non-concurrent forces is equal to the sum of the two forces and the angle between them.
10. The magnitude and direction of the resultant of a force system can be determined by drawing a free body diagram.

Types of Supports (Hinge, Roller) and Loads (Point, UDL, UVL)

- Supports are the elements of a structure which are used to keep it in place and transfer the load to the ground. They can be classified into two types:
 - Hinge: A hinge support allows one degree of freedom, i.e. rotation.
 - Roller: A roller support allows two degrees of freedom, i.e. rotation and translation.
- Loads are the forces which act on a structure and cause it to deform. They can be classified into three types:
 - Point Load: A point load is a concentrated force which acts on a single point on the structure.
 - UDL (Uniformly Distributed Load): A UDL is a force which is distributed over a certain length of the structure.
 - UVL (Uniformly Varying Load): A UVL is a force which varies in magnitude over a certain length of the structure.

Free Body Diagram for Unit 1 - Introduction to Mechanics in Fundamentals of Mechanical Engineering

- A free body diagram (FBD) is a graphical representation of the forces and moments that act on a body or system.
- It is used to represent the forces and moments acting on a body or system in order to analyze the motion of the body or system.
- An FBD should include all forces and moments that act on the body or system, including gravity, friction, air resistance, etc.
- The FBD should also include the reactions from the support surfaces, such as a wall or a floor.
- A free body diagram should be drawn in such a way that all forces and moments can be clearly seen, and that the direction of each force or moment can be determined.
- The FBD should be labeled with the type of force or moment (e.g. gravity, friction, etc.) and the magnitude of the force or moment (if known).
- When drawing a free body diagram, it is important to ensure that all forces

and moments are drawn to scale. This will help to accurately analyze the motion of the body or system.

Equilibrium Equations and Support Reactions

1. The **equilibrium equations** are a set of equations used to describe the forces and moments that act on a body in equilibrium. These equations are based on Newton's laws of motion and are used to determine the unknown forces and moments that act on a body.
2. A **support reaction** is the force or moment that a support exerts on a body in order to keep it in equilibrium. The support reaction can be determined by applying the equilibrium equations to the body.
3. The **free body diagram** is a graphical representation of the forces and moments that act on a body. It is used to help visualize the forces and moments that act on a body and to determine the support reactions.
4. The **equilibrium equations** can be written in vector form as:
 - $\sum F_x = 0$
 - $\sum F_y = 0$
 - $\sum M_z = 0$

Where F_x , F_y , and M_z are the x-component, y-component, and z-component of the force and moment vectors respectively.

5. The **support reactions** can be determined by solving the equilibrium equations for the unknown forces and moments. This can be done by using the methods of substitution or elimination.

Normal and Shear Stress

- Normal stress is a type of stress that is applied perpendicular to the surface of an object. It is caused by forces that are applied in the same direction.
- Shear stress is a type of stress that is applied parallel to the surface of an object. It is caused by forces that are applied in opposite directions.
- Both normal and shear stress can cause an object to deform.
- The magnitude of normal and shear stress is determined by the magnitude of the applied force and the area over which the force is applied.
- The stress experienced by an object is also affected by its material properties, such as its elasticity, strength, and ductility.
- The relationship between normal and shear stress is known as the Mohr-Coulomb failure criterion. This criterion states that the shear stress must exceed a certain threshold before the material fails.

Strain

- Strain is a measure of deformation that results from applied forces. It is defined as the relative change in length of a material when subjected to

an applied force.

- Strain can be described as either tensile or compressive. Tensile strain is a stretching of the material, while compressive strain is a squeezing of the material.
- Strain can also be described as either elastic or plastic. Elastic strain is a reversible deformation, while plastic strain is an irreversible deformation.
- Strain can be measured using various techniques, such as strain gauges and extensometers.
- Strain can be used to calculate the stress in a material, as well as the modulus of elasticity.

Hookes' Law

Hookes' law is a fundamental law of mechanics that states that the force required to extend or compress a spring is proportional to the distance the spring is extended or compressed. The law is named after Robert Hooke, who first proposed it in 1676.

In mathematical terms, Hookes' law states that the force (F) is proportional to the displacement (x) of a spring, according to the equation:

$$F = -kx$$

Where k is the spring constant, which is a measure of the stiffness of the spring.

Hookes' law is a useful tool for understanding the behavior of many mechanical systems, including pendulums, springs, and other objects that experience a restoring force proportional to the displacement. It is also useful for calculating the forces in compression and tension members, such as beams and columns.

Poisson's Ratio

- Poisson's ratio is a dimensionless number that describes the ratio of the transverse contraction strain to the axial extension strain in the direction of the applied load.
- It is denoted by the Greek symbol ν and is defined as the negative ratio of transverse strain ϵ_t to axial strain ϵ_a :

$$\nu = -\frac{\epsilon_t}{\epsilon_a}$$

- Poisson's ratio is a measure of the material's compressibility and is an important parameter in the determination of the elastic constants of a material.
- It is also used to calculate the stress-strain relationship in a material and is used in the design of mechanical components.
- The value of Poisson's ratio for most materials lies between 0 and 0.5, with most metals having a value of around 0.3.

Elastic Constants and Their Relationship

1. Elastic constants are physical quantities used to describe the mechanical behavior of a material when it is subjected to a force.
2. The most common elastic constants are Young's modulus (E), shear modulus (G), and Poisson's ratio (ν).
3. Young's modulus (E) is a measure of the stiffness of a material, and is equal to the ratio of the tensile stress to the strain.
4. Shear modulus (G) is a measure of the resistance of a material to shear stress. It is equal to the ratio of the shear stress to the shear strain.
5. Poisson's ratio (ν) is a measure of the lateral contraction of a material when it is subjected to a tensile force. It is equal to the ratio of the lateral strain to the longitudinal strain.
6. The relationship between the elastic constants is given by the following equation:

$$E = 2G(1 + \nu)$$

Stress-Strain Diagram for Ductile and Brittle Materials

- Stress-strain diagrams are graphical representations of the relationship between the stress applied to a material and the resulting strain.
- Ductile materials are able to undergo large amounts of deformation before failure, while brittle materials fail with relatively little deformation.
- The stress-strain diagram for ductile materials typically follows a curved line, while the diagram for brittle materials is usually linear.
- On a stress-strain diagram, the elastic limit is the point at which the material begins to deform permanently.
- The yield point is the point at which the material begins to deform plastically and the ultimate tensile strength is the maximum stress a material can withstand before failure.
- The strain hardening region is the area after the yield point where the material becomes stronger as it is deformed more.
- The ductile-to-brittle transition temperature is the temperature at which a ductile material becomes brittle.

Factor of Safety in Unit 1 - Introduction to Mechanics

- Factor of Safety (FoS) is an important concept in mechanical engineering, as it helps engineers ensure the safety and reliability of their designs.
- FoS is defined as the ratio of the ultimate strength of a material to the working stress it is subjected to.
- In order to calculate the FoS, engineers must consider the material properties, the geometry of the component, and the type of loading it will be subjected to.

- It is important to note that FoS is not a static value, and can change depending on the type of loading, the material properties, and the geometry of the component.
- FoS is typically expressed as a number greater than 1, with higher values indicating a higher margin of safety.
- When designing a component, engineers must consider the FoS and ensure that the component is designed to withstand the expected loading conditions.

Unit 2 - Introduction to IC Engines and Electric Vehicles

1. Internal Combustion Engines (ICE) are engines that burn fuel to generate power. They are used in a variety of applications, including automobiles, boats, aircraft, and other machinery.
2. ICEs can be powered by gasoline, diesel, or other fuels. They are typically more efficient than electric engines, but produce more emissions.
3. Electric Vehicles (EVs) are powered by electric motors and batteries. They are becoming increasingly popular due to their low emissions and high efficiency.
4. EVs are becoming more affordable and have a range of features that make them attractive to consumers. They are also becoming more widely available in many countries.
5. EV technology is constantly evolving, with new features and improvements being made all the time. This includes advances in battery technology, charging infrastructure, and autonomous driving technology.
6. ICEs and EVs both have advantages and disadvantages, and the decision of which type of vehicle to buy depends on individual needs and preferences.

IC Engines

- Internal combustion engines (IC engines) are heat engines in which combustion of a fuel occurs with an oxidizer (usually air) in a combustion chamber that is an integral part of the working fluid flow circuit.
- IC engines are the most commonly used form of heat engine and are found in a wide variety of applications, including automobiles, motorcycles, aircraft, ships, power plants, and stationary engines.
- IC engines are classified based on the type of fuel used, the number of cylinders, the type of ignition system, and the number of strokes per cycle.
- The four main types of IC engines are spark-ignition (SI) engines, compression-ignition (CI) engines, two-stroke engines, and four-stroke engines.
- SI engines use spark plugs to ignite the fuel-air mixture, while CI engines use compression of the air-fuel mixture to ignite the fuel.
- Two-stroke engines have two strokes per cycle and four-stroke engines have four strokes per cycle.
- The four strokes of a four-stroke engine are intake, compression, power,

and exhaust.

- The main components of an IC engine include the engine block, cylinder head, crankshaft, connecting rods, pistons, valves, fuel injectors, and spark plugs.
- The most common fuels used in IC engines are gasoline, diesel, and natural gas.
- The efficiency of an IC engine is affected by the compression ratio, the engine speed, the air-fuel ratio, and the exhaust gas temperature.

Basic Definition of Engine and Components

- An engine is a device that converts energy into useful work. It can be powered by a variety of sources, including electricity, steam, gas, or fuel.
- The components of an engine include the combustion chamber, valves, pistons, crankshaft, connecting rods, camshaft, intake and exhaust manifolds, and other components.
- The combustion chamber is the area in which the fuel and air are mixed and ignited.
- The valves control the flow of air and fuel into and out of the combustion chamber.
- The pistons convert the energy of the expanding gases into mechanical motion.
- The crankshaft converts the reciprocating motion of the pistons into rotary motion.
- The connecting rods link the crankshaft to the pistons.
- The camshaft is used to open and close the valves.
- The intake and exhaust manifolds direct the flow of air and fuel into and out of the combustion chamber.
- Other components, such as the fuel injectors, spark plugs, and cooling system, are also important in the operation of an engine.

Construction and Working of Two stroke and four stroke SI & CI engine

- Two-stroke engines are internal combustion engines that complete the entire four-stroke cycle in only two strokes of the piston. They are simpler and lighter than four-stroke engines and can produce more power for their size.
- Four-stroke engines are the most common type of internal combustion engine. They complete the four-stroke cycle in four piston strokes, which includes intake, compression, combustion, and exhaust. Four-stroke engines are more efficient than two-stroke engines and have a longer lifespan.
- The SI (Spark Ignition) engine is an internal combustion engine that uses a spark plug to ignite the fuel-air mixture. It is most commonly used in gasoline-powered vehicles.
- The CI (Compression Ignition) engine is an internal combustion engine

that uses compression to ignite the fuel-air mixture. It is most commonly used in diesel-powered vehicles.

- In a two-stroke SI engine, the intake, compression, combustion, and exhaust occur in two piston strokes. The intake stroke draws in a fuel-air mixture, which is then compressed in the compression stroke. The spark plug ignites the fuel-air mixture, and the combustion stroke pushes the piston down. The exhaust stroke then pushes out the exhaust gases.
- In a four-stroke SI engine, the intake, compression, combustion, and exhaust occur in four piston strokes. The intake stroke draws in a fuel-air mixture, which is then compressed in the compression stroke. The spark plug ignites the fuel-air mixture, and the combustion stroke pushes the piston down. The exhaust stroke then pushes out the exhaust gases.
- In a two-stroke CI engine, the intake, compression, combustion, and exhaust occur in two piston strokes. The intake stroke draws in a fuel-air mixture, which is then compressed in the compression stroke. The compression of the fuel-air mixture causes it to ignite, and the combustion stroke pushes the piston down. The exhaust stroke then pushes out the exhaust gases.
- In a four-stroke CI engine, the intake, compression, combustion, and exhaust occur in four piston strokes. The intake stroke draws in a fuel-air mixture, which is then compressed in the compression stroke. The compression of the fuel-air mixture causes it to ignite, and the combustion stroke pushes the piston down. The exhaust stroke then pushes out the exhaust gases.

Merits and Demerits for the Notes of the Unit 2 - Introduction to IC Engines and Electric Vehicles in the Subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

Merits

1. IC engines are more efficient than electric vehicles in terms of fuel economy.
2. IC engines have a longer life span than electric vehicles.
3. IC engines are more reliable and require less maintenance than electric vehicles.
4. IC engines are more powerful than electric vehicles.

Demerits

1. IC engines produce more emissions than electric vehicles.
2. IC engines are more expensive than electric vehicles.
3. IC engines require more frequent refuelling than electric vehicles.
4. IC engines are more susceptible to breakdowns than electric vehicles.

Scavenging Process for the Notes of Unit 2 - Introduction to IC Engines and Electric Vehicles in FUNDAMENTALS OF MECHANICAL ENGINEERING

1. Scavenging is the process of removing exhaust gases from the cylinders of an internal combustion engine.
2. It is important for the efficient operation of an engine as it allows for fresh air to be drawn into the cylinders for the combustion process.
3. The scavenging process is generally accomplished by a two-stroke cycle engine, which uses the piston to draw in the fresh air and push out the exhaust gases.
4. In a four-stroke cycle engine, the scavenging process is achieved by the use of a positive displacement blower or a turbocharger.
5. The scavenging process is also used in electric vehicles to remove the heat generated by the motor during operation.
6. The scavenging process helps to ensure that the motor does not overheat, which can lead to damage and reduced efficiency.
7. The scavenging process also helps to reduce emissions from the vehicle, as the exhaust gases are removed before they are released into the atmosphere.

Difference between Two-Stroke and Four-Stroke IC Engines and SI and CI Engines

1. Two-stroke engines complete a power cycle in two strokes of the piston, while four-stroke engines complete a power cycle in four strokes of the piston.
2. Two-stroke engines are simpler in design and have fewer moving parts than four-stroke engines.
3. Two-stroke engines typically have higher power-to-weight ratios than four-stroke engines, making them ideal for applications where weight is a concern.
4. Two-stroke engines are typically used in applications such as motorcycles, chainsaws, and outboard motors.
5. Four-stroke engines are typically used in cars, trucks, and other applications where fuel efficiency is a primary concern.
6. Spark-ignition (SI) engines use spark plugs to ignite a fuel-air mixture, while compression-ignition (CI) engines rely on the heat generated by compressing the fuel-air mixture to ignite it.
7. SI engines are typically used in applications such as cars, motorcycles, and lawn mowers, while CI engines are typically used in applications such as trucks, buses, and generators.

Electric Vehicles and Hybrid Vehicles

- Electric vehicles (EVs) are powered by electric motors and batteries, while hybrid vehicles (HVs) use a combination of electric motors and internal

combustion engines (ICEs).

- EVs are typically more energy efficient than HVs, as they have lower energy losses due to friction, air drag, and other factors.
- EVs have fewer moving parts than HVs, which reduces maintenance costs and increases reliability.
- EVs can be charged from an external power source, while HVs must be refueled with gasoline or diesel fuel.
- EVs can be powered by renewable energy sources, such as solar or wind power, while HVs are limited to non-renewable sources, such as petroleum-based fuels.
- EVs are typically quieter than HVs, as they have no engine noise.
- EVs have lower emissions than HVs, as they produce no direct emissions of pollutants.
- EVs can be more expensive than HVs, due to the cost of the electric motor and battery.

Components of an EV

1. Battery - The battery is the main source of power for an Electric Vehicle. It usually consists of multiple cells connected in series and parallel. The battery provides the necessary power to the motor to move the vehicle.
2. Motor - The motor is the main component of an Electric Vehicle. It is responsible for converting electrical energy from the battery into mechanical energy to move the vehicle.
3. Controller - The controller is responsible for controlling the flow of electricity from the battery to the motor. It is responsible for regulating the speed and power of the motor.
4. Charger - The charger is responsible for charging the battery of the Electric Vehicle. It converts the AC power from the wall outlet into DC power which is then used to charge the battery.
5. Inverter - The inverter is responsible for converting the DC power from the battery into AC power which is then used to power the motor.
6. Transmission - The transmission is responsible for transferring the mechanical power from the motor to the wheels of the vehicle. It is usually composed of a series of gears which are used to adjust the speed of the vehicle.

EV Batteries

- Electric vehicles (EVs) are powered by batteries that store electrical energy.
- Different types of batteries are used in EVs, including lead-acid, nickel-metal hydride, and lithium-ion.
- Lead-acid batteries are the oldest type of EV battery, but are heavy, bulky, and have a short range.
- Nickel-metal hydride batteries are lighter and more efficient than lead-acid

batteries, but still have a short range.

- Lithium-ion batteries are the most popular type of EV battery, as they are light, efficient, and have a long range.
- EV batteries are typically charged by plugging them into an external power source, but some vehicles are equipped with regenerative braking systems that can recharge the battery while driving.

Chargers for the Notes of the Unit 2 - Introduction to IC Engines and Electric Vehicles in the Subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

1. Internal combustion (IC) engines are used in vehicles to convert chemical energy into mechanical energy.
2. An electric vehicle (EV) is powered by one or more electric motors, which use electricity stored in batteries or fuel cells to power the vehicle.
3. Chargers are devices used to recharge the batteries in electric vehicles.
4. Chargers can be either AC or DC. AC chargers use alternating current to charge the battery, while DC chargers use direct current.
5. Chargers can be either Level 1, Level 2, or Level 3. Level 1 chargers use 120 volts, Level 2 chargers use 240 volts, and Level 3 chargers use 480 volts.
6. Chargers come in a variety of shapes and sizes, and can be installed either in the vehicle or in the home.
7. Charging times vary depending on the type of charger and the size of the battery.
8. Charging an EV can be more expensive than fueling an IC engine vehicle, but it is more efficient and produces fewer emissions.

Drives for the Notes of the Unit 2 - Introduction to IC Engines and Electric Vehicles in the Subject of Fundamentals of Mechanical Engineering

1. Internal Combustion (IC) Engines: IC engines are engines that convert chemical energy into mechanical energy. They are typically powered by gasoline, diesel, or other combustible fuels. IC engines are used in cars, trucks, boats, and other vehicles.
2. Electric Vehicles: Electric vehicles are powered by electric motors and powered by electricity from batteries, fuel cells, or other energy sources. They are typically more efficient than IC engines and have lower emissions.
3. Hybrid Vehicles: Hybrid vehicles are powered by both IC engines and electric motors. They are typically more efficient than either IC engines or electric vehicles alone.
4. Fuel Cells: Fuel cells are electrochemical devices that convert chemical energy into electrical energy. They are typically used in electric vehicles and are more efficient than IC engines or electric motors.

5. Thermodynamics: Thermodynamics is the study of energy and its conversion from one form to another. It is important to understand thermodynamics when studying IC engines and electric vehicles.

Transmission and Power Devices for Unit 2 - Introduction to IC Engines and Electric Vehicles

- Internal Combustion Engines (ICEs) are powered by a variety of fuel sources, including gasoline, diesel, and natural gas. The fuel is combusted inside the engine, creating power which is then used to move the vehicle.
- Electric Vehicles (EVs) are powered by electric motors which draw power from a battery or fuel cell. EVs are more efficient than ICEs and can be powered by renewable energy sources.
- Transmissions are used to transfer power from the engine to the wheels. Manual transmissions are operated by the driver, while automatic transmissions shift gears without the driver's input.
- Power devices are used to control the amount of power generated by the engine, and can include throttles, governors, and turbochargers.

Advantages and Disadvantages of EVs

Advantages: - Low cost of operation due to lower fuel costs. - Low emissions and no need for exhaust systems. - Quieter operation than traditional internal combustion engines. - Improved acceleration and torque compared to internal combustion engines. - Improved range and charging times due to advances in battery technology.

Disadvantages: - High initial costs due to the cost of the battery. - Limited range compared to traditional internal combustion engines. - Longer charging times than traditional internal combustion engines. - Limited availability of charging points in some areas. - High maintenance costs due to the complexity of the technology.

Hybrid Electric Vehicles

Hybrid electric vehicles (HEVs) are a type of vehicle that combines the traditional internal combustion engine (ICE) with an electric motor. They are designed to reduce fuel consumption and emissions while providing a more efficient driving experience.

HEVs are typically powered by a combination of gasoline and electricity, but may also include diesel, natural gas, and other fuels. The electric motor is usually powered by a rechargeable battery, which can be recharged from a wall outlet or from regenerative braking. When the battery is depleted, the ICE will kick in to provide power.

HEVs offer several advantages over traditional ICE vehicles. They are more efficient, produce fewer emissions, and are generally quieter. Additionally, they

can provide a smoother and more comfortable ride.

HEVs are becoming increasingly popular due to their efficiency and environmental benefits. They are also becoming more affordable, making them a viable option for many drivers.

HEVs are an important part of the future of transportation, and their popularity is only likely to increase in the coming years.

HEV Drive Train Components

1. Hybrid Electric Vehicle (HEV): An HEV is a vehicle that combines an internal combustion engine (ICE) with an electric motor. The electric motor is powered by a battery, which is recharged by the ICE.
2. Battery: The battery stores energy from the ICE and supplies power to the electric motor.
3. Electric Motor: The electric motor is used to propel the vehicle and is powered by the battery.
4. Internal Combustion Engine (ICE): The ICE is used to generate power for the electric motor and to recharge the battery.
5. Regulator/Inverter: The regulator/inverter is used to control the flow of electricity between the battery and the electric motor.
6. Transmission: The transmission is used to transfer power from the ICE to the wheels.

Advantages of Hybrid Vehicles (HV)

1. Hybrid vehicles use both an internal combustion engine (ICE) and an electric motor. This combination of both technologies allows the vehicle to be more efficient and reduce emissions.
2. Hybrid vehicles can switch between the ICE and the electric motor depending on the driving conditions. This allows for better fuel economy and improved performance.
3. Hybrid vehicles are more reliable than traditional ICE vehicles due to their ability to switch between power sources. This reduces the risk of breakdowns and increases the lifespan of the vehicle.
4. Hybrid vehicles are more environmentally friendly than traditional ICE vehicles due to their lower emissions. This improves air quality and helps reduce the effects of global warming.
5. Hybrid vehicles are becoming increasingly popular due to their efficiency and environmental benefits. This allows for more choice when selecting a new vehicle.

Unit 3 - Introduction to Refrigeration and Air-Conditioning

1. Refrigeration is the process of cooling a space, substance, or system to lower and maintain its temperature below the ambient one.
2. Air-conditioning is a type of refrigeration technology that uses mechanical, electrical, and chemical processes to cool air or remove moisture from it.
3. Refrigeration and air-conditioning systems are used to control temperature and humidity levels in residential, commercial, and industrial buildings.
4. Refrigerants are the fluids used in refrigeration and air-conditioning systems to absorb and release heat.
5. Refrigeration and air-conditioning systems can be divided into two categories: vapor-compression and absorption systems.
6. Vapor-compression systems are the most common type of refrigeration and air-conditioning systems and use a compressor to increase the pressure of the refrigerant.
7. Absorption systems use a chemical process to absorb heat from the air or space being cooled.
8. Refrigeration and air-conditioning systems are used in a variety of applications, including food storage and preservation, medical care, and industrial processes.

Refrigeration for the notes of the Unit 3 - Introduction to Refrigeration and Air-Conditioning in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

1. Refrigeration is the process of removing heat from a space, object or substance in order to lower its temperature.
2. Air-conditioning is the process of controlling the temperature, humidity, air circulation and air cleanliness of an environment.
3. Refrigeration and air-conditioning systems are used to maintain comfortable temperatures in buildings, transportation vehicles and storage facilities.
4. Refrigeration and air-conditioning systems use a variety of components, including compressors, evaporators, condensers and expansion devices.
5. Refrigerants are substances that are used to transfer heat from one location to another.
6. The efficiency of a refrigeration or air-conditioning system is determined by the type of refrigerant used, the size of the system and the amount of heat it must remove.
7. Proper maintenance and servicing of a refrigeration or air-conditioning system is essential for its efficient operation.

Refrigerating Effect

- Refrigeration is the process of cooling a space, substance, or system by removing heat from it. This is usually done by transferring the heat to a colder environment.
- Refrigeration is used in many applications, including food storage, air conditioning, and industrial cooling.
- Air-conditioning is a form of refrigeration that uses a compressor, evaporator, and condenser to cool air.
- The refrigerating effect is the amount of heat that must be removed from a system in order to achieve a desired temperature.
- This effect is measured in British Thermal Units (BTUs) and is typically expressed in terms of the amount of heat removed per hour.
- The refrigerating effect is affected by the type of refrigerant used, the size of the system, and the temperature of the environment.

Ton of Refrigeration

1. The ton of refrigeration (TR) is a unit of measurement used to measure cooling capacity. It is the amount of cooling a system can provide in an hour.
2. One ton of refrigeration is equal to 12,000 BTU/hr (British Thermal Units per hour).
3. Refrigeration systems are used to cool air or other substances, such as food.
4. Refrigeration systems are used in air conditioning, food processing, and other applications.
5. Refrigeration systems use a refrigerant to absorb heat from the air or substance being cooled and transfer it to the outside air.
6. Refrigeration systems use compressors, condensers, evaporators, and expansion valves to transfer heat.
7. The efficiency of a refrigeration system is measured by its coefficient of performance (COP).
8. The COP is the ratio of the cooling capacity of the system to the energy input.
9. The higher the COP, the more efficient the system is.

Coefficient of Performance for Introduction to Refrigeration and Air-Conditioning

1. The coefficient of performance (COP) is a measure of the efficiency of a refrigeration system or air conditioning system. It is defined as the ratio of the cooling capacity of the system to the power input.
2. The COP can be calculated using the following formula:
$$\text{COP} = \frac{\text{Cooling Capacity}}{\text{Power Input}}$$

3. The higher the COP, the more efficient the system is. Generally, a COP of 3 or higher is considered to be very efficient.
4. The COP of a system can be improved by increasing the efficiency of the compressor, improving the insulation of the system, and using a higher pressure refrigerant.
5. The COP of a system can also be affected by the ambient temperature, the size of the system, and the type of refrigerant used.

Methods of Refrigeration

1. Vapor Compression Refrigeration: This is the most commonly used method of refrigeration and involves the use of a compressor to compress a refrigerant vapor. This increases the pressure and temperature of the refrigerant, which then flows into a condenser where it is cooled and condensed into a liquid. The liquid is then passed through an expansion valve, which reduces its pressure and temperature, and then into the evaporator. In the evaporator, the liquid refrigerant absorbs heat from the air or other substance to be cooled and evaporates into a vapor. The vapor is then drawn back into the compressor and the cycle is repeated.
2. Absorption Refrigeration: This method of refrigeration uses a heat source to drive the refrigeration cycle. In this process, a liquid refrigerant is absorbed by another liquid, usually water or ammonia, which is then evaporated in the evaporator. The vapor is then condensed in the condenser and the liquid refrigerant is then returned to the evaporator.
3. Thermoelectric Refrigeration: This method of refrigeration utilizes the Peltier effect to produce cooling. In this process, a current is passed through two dissimilar materials, creating a temperature difference between the two. This temperature difference is then used to cool a substance.
4. Magnetic Refrigeration: This method of refrigeration utilizes magnetic materials to produce cooling. In this process, a magnetic material is subjected to a changing magnetic field, which causes it to heat up. The heat is then used to cool a substance.

Construction and Working of Domestic Refrigerator

1. Refrigerators are cooling appliances that use a refrigeration cycle to cool the air inside the refrigerator.
2. The main components of a refrigerator are the compressor, condenser, evaporator, and expansion valve.
3. The compressor compresses the refrigerant gas, which causes it to become hot and high pressure.
4. The hot and high-pressure gas then enters the condenser, where it is cooled and turned into a liquid.

5. The liquid then passes through the expansion valve, where it is turned into a low-pressure and low-temperature gas.
6. The gas then enters the evaporator, where it absorbs heat from the air inside the refrigerator and turns back into a liquid.
7. The liquid then passes back through the compressor, completing the cycle and cooling the air inside the refrigerator.

Concept of Heat Pump

1. A heat pump is a device that uses mechanical energy to transfer thermal energy from a low-temperature source to a high-temperature sink.
2. Heat pumps are often used for heating and cooling applications, such as air conditioners and refrigerators.
3. Heat pumps work by compressing and expanding a refrigerant, which changes its temperature.
4. The heat pump is a closed-loop system, meaning that the same refrigerant is used over and over again.
5. Heat pumps can be used to transfer heat from a cool area to a warm area, or vice versa.
6. Heat pumps are more efficient than traditional heating and cooling systems, as they use less energy to transfer the same amount of heat.
7. Heat pumps are also more environmentally friendly than traditional systems, as they do not produce any greenhouse gases.
8. Heat pumps can be used for both heating and cooling applications, making them a versatile and cost-effective solution for many different types of buildings.

Air-Conditioning for the notes of the Unit 3 - Introduction to Refrigeration and Air-Conditioning in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

- Air-conditioning is a process of controlling the temperature, humidity, and purity of the air in an enclosed space.
- It is achieved by using a refrigeration cycle and a heating cycle, both of which are controlled by a thermostat.
- The main components of an air-conditioning system are the compressor, condenser, evaporator, and expansion valve.
- The compressor compresses the refrigerant, which increases its temperature and pressure.
- The condenser then cools the refrigerant, which decreases its temperature and pressure.
- The cooled refrigerant is then passed through the evaporator, which absorbs the heat from the air in the enclosed space, thus cooling it.
- The expansion valve then reduces the pressure of the refrigerant, allowing it to evaporate and absorb more heat from the air.
- The cooled air is then circulated through the space by a fan.

- The thermostat is used to regulate the temperature of the space, by controlling the compressor and the heater.

Its Meaning and Application for the Notes of the Unit 3 - Introduction to Refrigeration and Air-Conditioning in the Subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

1. Refrigeration and air-conditioning are two closely related fields of engineering. Refrigeration involves the transfer of heat from one area to another, while air-conditioning is the process of controlling the temperature, humidity, and air quality of a given space.
2. Refrigeration is used in a variety of applications, from food storage and preservation to industrial cooling and air-conditioning. It is also used in residential and commercial buildings to maintain a comfortable environment.
3. Air-conditioning is used to control the temperature, humidity, and air quality of a given space. It is used in homes, offices, and other types of buildings to maintain a comfortable environment.
4. The principles of refrigeration and air-conditioning are based on thermodynamics, the study of heat and energy. The principles involve the transfer of heat from one area to another, and the control of temperature, humidity, and air quality.
5. Refrigeration and air-conditioning systems are composed of several components, such as compressors, condensers, evaporators, and expansion valves. These components work together to transfer heat from one area to another and to control the temperature, humidity, and air quality of a space.
6. Refrigeration and air-conditioning systems are designed to be efficient and reliable. They are also designed to be safe and to minimize the environmental impact of their operation.

Humidity

Humidity is an important factor to consider when dealing with refrigeration and air-conditioning systems. It is the amount of water vapor in the air and is measured in relative humidity (RH). High relative humidity can cause condensation on the evaporator coils and reduce the efficiency of the system.

Humidity can also cause corrosion and mold growth, which can be damaging to the system. To prevent these issues, the humidity should be maintained at a low level. This can be done by using a dehumidifier, a humidifier, or an air conditioner.

Humidity can also affect the comfort of occupants in a room. High humidity can make the air feel warmer, while low humidity can make the air feel cooler. Therefore, it is important to maintain a comfortable level of humidity in a space.

Humidity also affects the performance of air-conditioning systems. At higher humidity levels, the system will need to work harder to cool the air, which can lead to increased energy consumption. Therefore, it is important to maintain the humidity levels in a space to ensure the efficiency of the system.

Dry Bulb

- The dry bulb temperature is the temperature of the air as measured by a thermometer exposed to the air, but shielded from radiation and moisture.
- It is the most common form of measurement used in meteorology and HVAC applications.
- In HVAC applications, the dry bulb temperature is used to calculate the humidity ratio and the dew point temperature.
- The dry bulb temperature is also used to calculate the sensible heat load, which is the amount of heat energy that needs to be added or removed from a space to maintain a comfortable environment.
- The dry bulb temperature can be measured with a thermometer, or it can be estimated by measuring the wet bulb temperature and then subtracting the wet bulb depression.

Wet Bulb

- The wet bulb temperature is the temperature that is read by a thermometer when it is covered with a wet cloth or wick.
- The wet bulb temperature is lower than the dry bulb temperature because the evaporation of water from the wet cloth or wick causes cooling.
- The wet bulb temperature is used to measure the humidity of the air.
- In air-conditioning and refrigeration, the wet bulb temperature is used to calculate the cooling capacity of the system.
- The wet bulb temperature is also used to calculate the dew point temperature, which is the temperature at which the air is saturated with water vapor.

Dew Point Temperatures for Unit 3 - Introduction to Refrigeration and Air-Conditioning

- Dew point temperature is the temperature at which air becomes saturated with water vapor.
- When air is cooled to its dew point temperature, the air can no longer hold all of the water vapor and some of the water vapor condenses into liquid droplets.
- The dew point temperature is an important factor in air conditioning and refrigeration systems as it determines the amount of moisture in the air.
- The dew point temperature is also used to calculate the relative humidity of the air.
- The dew point temperature is affected by the temperature and pressure of the air, as well as the amount of water vapor present in the air.

Comfort Conditions for Unit 3 - Introduction to Refrigeration and Air-Conditioning in FUNDAMENTALS OF MECHANICAL ENGINEERING

- Comfort conditions are defined as the desired environment for human comfort, which is usually a temperature range between 18-25 degrees Celsius and a relative humidity between 40-60%.
- Refrigeration and air-conditioning systems are used to create these comfort conditions.
- Refrigeration systems use a refrigerant to absorb heat from the environment and transfer it to another location, thus cooling the environment.
- Air-conditioning systems use a refrigerant to absorb heat from the environment and transfer it to an outdoor unit, thus cooling the environment.
- The temperature and humidity of the environment can be controlled using a thermostat, which is a device that senses the temperature and humidity of the environment and adjusts the cooling system accordingly.
- Proper maintenance and servicing of the refrigeration and air-conditioning systems is essential to ensure that the comfort conditions are maintained.

Construction and Working of Window Air Conditioner

1. Window air conditioners are self-contained units that are usually installed in a window or wall opening.
2. The main components of a window air conditioner are the compressor, condenser, expansion valve, and evaporator.
3. The compressor compresses the refrigerant gas, which increases its temperature and pressure.
4. The hot, high-pressure gas then flows through the condenser, where it is cooled and converted back into a liquid.
5. The liquid then passes through the expansion valve and is evaporated in the evaporator.
6. The evaporator absorbs heat from the air in the room, cooling it down.
7. The cooled air is then blown back into the room.
8. The refrigerant gas then returns to the compressor, where the cycle begins again.

Unit 4 - Introduction to Fluid Mechanics and Applications

1. Fluid Mechanics is the study of the behavior of liquids and gases when subjected to forces and changes in temperature and pressure.
2. Fluid dynamics is the branch of fluid mechanics that deals with the motion of fluids.
3. Fluid statics is the branch of fluid mechanics that deals with the behavior of fluids at rest.
4. Bernoulli's equation is an important equation in fluid mechanics that relates pressure, velocity, and elevation of a fluid.

5. The Navier-Stokes equations are a set of equations used to describe the motion of a fluid in a given region.
6. Applications of fluid mechanics include aerodynamics, hydrodynamics, and fluid power.
7. Aerodynamics is the study of the motion of air and other gases.
8. Hydrodynamics is the study of the motion of liquids.
9. Fluid power is the use of fluids to generate mechanical energy, such as in hydraulic systems.

Introduction to Fluid Mechanics and Applications

1. Fluid mechanics is a branch of physics that deals with the behavior of fluids (liquids and gases) and their interactions with surfaces.
2. It is a sub-discipline of mechanics, which is the study of the effects of forces on objects.
3. Fluid mechanics is an important part of mechanical engineering, as it is used to understand and design systems involving liquids and gases.
4. Fluid mechanics is used to analyze and design a wide range of systems, from aircraft to water treatment plants.
5. In fluid mechanics, different properties of fluids such as pressure, velocity, and density are studied.
6. Fluid dynamics is the study of the motion of fluids and their effects on solid objects.
7. It involves the study of the forces that act on a fluid and the behavior of the fluid in response to those forces.
8. Applications of fluid mechanics include aircraft design, the design of water treatment plants, and the design of pipelines and other systems involving the transport of fluids.

Fluids Properties for the Notes of the Unit 4 - Introduction to Fluid Mechanics and Applications in the Subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

- Fluids are substances that are capable of flowing and are not rigid. Examples of fluids include liquids and gases.
- Fluids can be characterized by their density, viscosity, and surface tension.
- Density is the mass per unit volume of a substance and is usually expressed in kg/m^3 .
- Viscosity is a measure of a fluid's resistance to flow and is usually expressed in $\text{Pa}\cdot\text{s}$.
- Surface tension is a measure of the cohesive forces between molecules of a fluid and is usually expressed in N/m .
- Fluids can also be classified as either Newtonian or non-Newtonian, depending on their behavior under stress. Newtonian fluids have a constant viscosity, while non-Newtonian fluids have a variable viscosity.
- The study of fluid mechanics deals with the behavior of fluids, including

the forces acting on them, and the resulting motion of the fluids.

- Applications of fluid mechanics include aerodynamics, hydrodynamics, and thermodynamics.

Pressure

1. Pressure is a measure of the force per unit area applied to a surface. It is a scalar quantity and is measured in pascals (Pa).
2. Pressure can be caused by a variety of factors, including the weight of a fluid, the force of gravity, or the movement of a fluid.
3. Pressure is a key factor in many engineering applications, such as hydraulics and pneumatics.
4. Pressure can be calculated by multiplying the density of a fluid by its acceleration due to gravity and the height of the fluid. This is known as the hydrostatic equation.
5. Pressure can also be calculated using Bernoulli's equation, which is a statement of the conservation of energy in a fluid.
6. Pressure can also be measured using a manometer, which is a device that measures the difference in pressure between two points.
7. Pressure is an important factor in fluid mechanics, as it can be used to calculate the forces acting on a fluid, the flow rate of a fluid, and the energy of a fluid.
8. Pressure is also used to calculate the lift and drag forces acting on an object in a fluid, as well as the pressure drop across a pipe.

Density for Unit 4 - Introduction to Fluid Mechanics and Applications in Fundamentals of Mechanical Engineering

- Density is a measure of mass per unit volume. It is an intrinsic physical property of a material and is usually expressed in kilograms per cubic meter (kg/m^3).
- Density is an important factor in determining the behavior of fluids, as it affects the speed of sound, pressure, and other properties.
- The density of a fluid is determined by its composition, temperature, and pressure.
- Density is an important factor in determining the stability of a fluid, as well as its ability to mix with other fluids.
- In fluid mechanics, density is used to calculate the force of gravity, buoyancy, and other forces.
- Density is also used to calculate the Reynolds number, which is a measure of the relative importance of inertia and viscous forces in a flow.
- Density is also used to calculate the pressure drop in a pipe or channel, and the friction factor in a pipe.

Dynamic and Kinematic Viscosity

- Dynamic viscosity is a measure of a fluid's resistance to shear or flow. It is also known as the absolute viscosity and is measured in units of Pascal-seconds (Pa-s).
- Kinematic viscosity is the ratio of the dynamic viscosity to the density of a fluid. It is measured in units of square meters per second (m^2/s).
- The viscosity of a fluid is a measure of its resistance to flow. Viscosity is an important property of fluids because it affects the behavior of the fluid in a variety of ways.
- Viscosity affects the drag force acting on a body moving through a fluid, the rate of heat transfer, and the rate of mass transfer.
- Viscosity also affects the rate of diffusion of a substance in a fluid, and the rate of mixing of two fluids.
- Viscosity is an important factor in determining the flow of fluids in pipes and other channels, and the flow of fluids over surfaces.
- The viscosity of a fluid is determined by the temperature, pressure, and composition of the fluid. The viscosity of a fluid increases with increasing temperature, and decreases with increasing pressure.

Specific Gravity

Specific gravity is a measure of the relative density of a substance compared to water. It is defined as the ratio of the density of a substance to the density of water at a given temperature.

The specific gravity of a substance can be used to determine the composition of a mixture, as different substances have different specific gravities. For example, the specific gravity of a mixture of oil and water can be used to determine the percentage of oil in the mixture.

Specific gravity is also used to calculate the buoyancy of a substance. The greater the specific gravity, the greater the buoyancy. This is because higher specific gravity substances are denser and therefore more likely to sink.

Specific gravity is also used in the field of mechanical engineering to calculate the pressure at a certain depth in a fluid. This is because the pressure at a certain depth is proportional to the density of the fluid at that depth. Therefore, the pressure at a certain depth can be calculated by multiplying the specific gravity of the fluid by the depth.

Newtonian and Non-Newtonian Fluids

1. Newtonian fluids are fluids that obey the linear relationship between shear stress and shear rate. This means that the shear rate of the fluid is directly proportional to the shear stress applied to it. Examples of Newtonian fluids include water, air, and most common liquids.

2. Non-Newtonian fluids are fluids that do not obey the linear relationship between shear stress and shear rate. This means that the shear rate of the fluid is not directly proportional to the shear stress applied to it. Examples of Non-Newtonian fluids include ketchup, toothpaste, and paint.
3. Newtonian fluids are used in a variety of applications, such as lubrication, hydraulics, and cooling systems. Non-Newtonian fluids are used in applications such as adhesives and paints.
4. The viscosity of a Newtonian fluid is constant and independent of the shear rate. The viscosity of a Non-Newtonian fluid is not constant and can vary greatly depending on the shear rate.
5. The flow of a Newtonian fluid is laminar, meaning that the flow is smooth and the particles move in a straight line. The flow of a Non-Newtonian fluid is turbulent, meaning that the flow is chaotic and the particles move in a random pattern.

Pascal's Law and Continuity Equation

Pascal's Law states that when a pressure is applied to a confined fluid, the pressure is transmitted undiminished in all directions. This law is used to explain the behavior of liquids when confined in a container.

The Continuity Equation states that the rate of flow of a fluid through a given cross-sectional area is equal to the product of the area and the average velocity of the fluid. This equation is used to describe the behavior of fluids in various applications, such as fluid flow through pipes or channels.

These two laws are important for understanding the behavior of fluids and are essential for engineers in the field of mechanical engineering. By understanding these laws, engineers can better design systems and components that involve the use of fluids.

Working principles of hydraulic turbines (Pelton Wheel and Francis) & pumps (Centrifugal and Reciprocating) and their classifications and hydraulic lift for the notes of the Unit 4 - Introduction to Fluid Mechanics and Applications in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

- Hydraulic turbines are machines that convert the potential energy of a falling water stream into mechanical energy. The two most common types of hydraulic turbines are the Pelton wheel and the Francis turbine.
- The Pelton wheel is a type of impulse turbine that consists of a series of buckets mounted around the circumference of a wheel. The water is directed onto the buckets at a high velocity, causing them to rotate and converting the energy of the water into mechanical energy.

- The Francis turbine is a type of reaction turbine that consists of a series of curved blades mounted around the circumference of a wheel. The water is directed onto the blades at a high velocity, causing them to rotate and converting the energy of the water into mechanical energy.
- Pumps are machines that convert mechanical energy into the potential energy of a liquid. The two most common types of pumps are the centrifugal pump and the reciprocating pump.
- The centrifugal pump is a type of rotodynamic pump that consists of an impeller mounted on a shaft. The impeller is driven by a motor and rotates, causing the liquid to be drawn in and then ejected at a high velocity.
- The reciprocating pump is a type of positive displacement pump that consists of a piston mounted on a cylinder. The piston is driven by a motor and reciprocates, causing the liquid to be drawn in and then ejected at a high pressure.
- Hydraulic lifts are machines that convert the potential energy of a liquid into mechanical energy. The hydraulic lift consists of a cylinder filled with a liquid, such as oil or water, and a piston. When the piston is pushed down, the liquid is forced out of the cylinder, causing the lift to move upwards. When the piston is pulled up, the liquid is drawn back into the cylinder, causing the lift to move downwards.

Unit 5 - Introduction to Measurement and Mechatronics

- Measurement is the process of determining the size, amount, or degree of something. It is used in a variety of fields, including engineering, physics, chemistry, and mathematics.
- Mechatronics is the combination of mechanics and electronics. It is a field of engineering that combines mechanical, electrical, and computer engineering to create systems that are more efficient and reliable.
- Measurement systems are used to measure physical quantities, such as temperature, pressure, and force. They can also be used to measure abstract quantities, such as time, speed, and distance.
- Mechatronic systems are used to control and monitor mechanical processes. They can be used to control robots, industrial machines, and other automated systems.
- Measurement and mechatronic systems are used in a variety of industries, including automotive, aerospace, medical, and manufacturing. They are also used in research and development.

Introduction to Measurement

- Measurement is the process of determining the size, amount, or degree of something. It is a fundamental concept in engineering, as it is used to quantify physical properties and to compare objects.

- Measurement is used to quantify physical properties such as length, mass, time, temperature, and pressure. It is also used to compare objects, such as comparing the size of a car to that of a truck.
- In engineering, measurements are typically made with instruments such as calipers, rulers, scales, and thermometers. These instruments are used to measure a variety of physical properties, such as length, weight, temperature, pressure, and speed.
- Measurement is also used to compare objects, such as comparing the size of a car to that of a truck. This comparison can be made using a variety of methods, such as using a ruler to measure the length of the car and the truck, or using a scale to measure their weights.
- Measurement is also used to calculate the amount of energy, force, or momentum in a system. This calculation is typically done using equations, such as the equation for the force of gravity.
- Mechatronics is the combination of mechanical and electronic engineering. It is used to design and build machines that combine mechanical and electronic components. Mechatronics is used in a variety of applications, such as robotics, automation, and control systems.
- Mechatronics is used to measure physical properties, such as length, mass, time, temperature, and pressure. It is also used to compare objects, such as comparing the size of a car to that of a truck. Mechatronics is also used to calculate the amount of energy, force, or momentum in a system.

Concept of Measurement

1. Measurement is the process of determining the size, amount, or degree of something. It is used to compare an unknown quantity to a known quantity.
2. Measurement systems are used to measure physical quantities such as length, area, volume, weight, temperature, pressure, and force.
3. Measurement systems can also be used to measure intangible quantities such as time, velocity, acceleration, power, and energy.
4. Measurement systems are used in a variety of fields, including engineering, science, and industry.
5. Mechatronics is a field of engineering that combines mechanical, electrical, and computer engineering to create systems that are capable of sensing, controlling, and communicating with the environment.
6. Mechatronic systems are used in a variety of applications, such as robotics, automation, and control systems.
7. Introduction to Measurement and Mechatronics is a course that provides an overview of the fundamentals of measurement systems and mechatronic systems.
8. The course covers topics such as sensors, transducers, signal conditioning, data acquisition, and control systems.

Error in Measurements

1. Errors in measurements are inevitable due to various factors.
2. Sources of error can be divided into two categories: random and systematic.
3. Random errors are caused by unpredictable fluctuations in the measurement process, and can be minimized by taking multiple measurements and calculating the average.
4. Systematic errors are caused by systematic errors in the measurement process, such as incorrect calibration of the measuring instrument, or incorrect use of the instrument.
5. To reduce systematic errors, it is important to use properly calibrated measuring instruments and to use them correctly.
6. In addition, it is important to be aware of the limitations of the measuring instrument and to take into account any potential sources of error when interpreting the results.

Calibration for the notes of the Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

1. Calibration is a process used to adjust a measuring instrument or system so that its readings are accurate and consistent.
2. Calibration is essential for ensuring the accuracy of measurements and is an important part of quality control.
3. It is important to understand the basic principles of measurement and mechatronics in order to properly calibrate a system.
4. Calibration involves setting the input and output values of a system to known or accepted standards.
5. It is important to use the correct calibration equipment and procedures in order to ensure accuracy.
6. The accuracy of a calibration can be affected by environmental factors such as temperature, humidity, and vibration.
7. Calibration should be performed regularly to ensure the accuracy of the system.
8. Proper calibration can help to reduce errors and improve the accuracy of measurements.

Measurements of Pressure (Bourdon Tube Pressure and U-Tube Manometer)

- Bourdon Tube Pressure: This type of pressure measurement is based on the principle of elasticity. It is a curved tube filled with a fluid, usually oil or water. When pressure is applied to the tube, it straightens and causes the pointer to move on a scale.
- U-Tube Manometer: This type of pressure measurement is based on the

principle of hydrostatic pressure. It consists of two vertical tubes connected at the bottom, with one side being open to the atmosphere. When pressure is applied to the open side, the fluid in the tubes rises and falls, and the difference in height can be measured by the pointer on the scale.

Temperature (Thermocouple and Optical Pyrometer)

1. A thermocouple is a device that measures temperature by using two different metals joined together. The junction of the two metals produces a voltage that is proportional to the temperature.
2. An optical pyrometer is a device that measures temperature by using the thermal radiation emitted by an object. The intensity of the radiation is proportional to the temperature of the object.
3. Thermocouples are often used in industrial applications because they are relatively inexpensive and easy to use. They can measure temperatures up to 2000°C.
4. Optical pyrometers are often used in high-temperature applications because they are more accurate than thermocouples. They can measure temperatures up to 3500°C.
5. Both thermocouples and optical pyrometers are used in the field of mechatronics to measure temperature in a variety of applications.

Mass Flow Rate (Venturi Meter and Orifice Meter)

1. Mass flow rate is the rate of mass flow of a fluid through a given cross-sectional area. It can be measured using a variety of methods such as Venturi meters and orifice meters.
2. Venturi meters are devices that measure the mass flow rate of a fluid by measuring the pressure drop across a constriction in a pipe. The pressure drop is proportional to the mass flow rate of the fluid.
3. Orifice meters are devices that measure the mass flow rate of a fluid by measuring the pressure drop across an orifice plate. The pressure drop is proportional to the mass flow rate of the fluid.
4. Both Venturi meters and orifice meters are widely used in industry to measure the mass flow rate of fluids. They are particularly useful for measuring the mass flow rate of gases, as they are more accurate than other methods such as Coriolis meters.
5. It is important to note that the accuracy of the measurements obtained from these devices depends on the accuracy of the calibration of the device. Improper calibration can lead to inaccurate measurements, so it is important to ensure that the device is properly calibrated before use.

Strain (Bonded and Unbonded Strain Gauge)

Strain is a measure of the deformation of a material due to an applied force. Strain gauges are devices used to measure strain in a material. They can be either bonded or unbonded.

Bonded Strain Gauge

A bonded strain gauge is a device that is directly attached to the material being tested. It is made up of a thin metal foil, which is attached to the material using an adhesive. The strain gauge is then connected to a measuring device, such as a strain gauge amplifier, which measures the change in resistance of the foil as the material is strained.

Unbonded Strain Gauge

An unbonded strain gauge is a device that is not attached directly to the material being tested. Instead, it is attached to a support structure, such as a steel frame, and the strain in the material is measured indirectly by measuring the strain in the support structure. This type of strain gauge is often used in applications where the material being tested cannot be directly accessed, such as in pipes and tanks.

Force (Proving Ring)

- Proving ring is a device used to measure the force applied to an object.
- It is a type of dynamometer, which is a device used to measure forces and torques.
- The proving ring works by applying a known force to an object and measuring the resulting displacement or deformation.
- Proving rings are used in a variety of applications, such as testing the strength of materials, measuring the force of a motor, and calibrating other force measuring instruments.

Torques (Prony Brake Dynamometer)

- Prony brake dynamometers are devices used to measure the torque applied to an object.
- They work by applying a known torque to an object and measuring the resulting displacement or deformation.
- Prony brake dynamometers are used in a variety of applications, such as testing the strength of materials, measuring the torque of a motor, and calibrating other torque measuring instruments.
- Prony brake dynamometers are also commonly used in automotive and aerospace applications.

Concepts of Accuracy for the Notes of the Unit 5 - Introduction to Measurement and Mechatronics in the Subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

1. Accuracy is the degree of conformity of a measured value to a standard or known value.
2. Accuracy is usually expressed as a percentage of error, which is the difference between the measured value and the standard value.
3. Resolution is the smallest change in the measured value that can be detected by the measuring system.
4. Repeatability is the ability of a measuring system to reproduce the same result when the same measurement is repeated under the same conditions.
5. Precision is the degree of reproducibility of a measuring system, which is determined by the number of significant figures in a measured value.
6. Linearity is the ability of a measuring system to produce a linear response over its entire range of measurement.
7. Hysteresis is the difference in the measured value when the same measurement is made in two different directions.
8. Drift is the gradual change in the measured value over time due to changes in the environment or the measuring system.
9. Calibration is the process of adjusting a measuring system to a known standard.

Precision and Resolution

- Precision and resolution are two important concepts in measurement and mechatronics.
- Precision refers to the degree of accuracy of a measurement. It is the degree to which repeated measurements of the same quantity will give the same result.
- Resolution is the smallest change in a quantity that can be detected by a measuring device. It is the smallest difference between two values that can be distinguished by a measuring device.
- Precision and resolution are closely related. Higher precision implies higher resolution.
- Precision and resolution are important factors to consider when selecting a measuring device.

Introduction to Mechatronic Systems

1. Mechatronic systems are a combination of mechanical, electrical, and computer engineering components that are used to create automated systems.
2. These systems are used in a wide range of applications, from controlling robots to creating automated manufacturing processes.
3. Mechatronic systems are designed to be reliable and efficient, and can be used to increase productivity and accuracy in manufacturing processes.

4. The components of a mechatronic system include sensors, actuators, controllers, and communication systems.
5. Sensors are used to measure physical parameters such as temperature, pressure, and position.
6. Actuators are used to convert electrical signals into mechanical movement, such as motors and solenoids.
7. Controllers are used to process the data from the sensors and to send signals to the actuators to control the system.
8. Communication systems are used to transmit data between the components of the system.
9. Mechatronic systems are used in a variety of applications, from controlling robots to creating automated manufacturing processes.
10. Mechatronic systems can be used to increase productivity and accuracy in manufacturing processes.

Evolution of Measurement and Mechatronics

1. Measurement is the process of obtaining a numerical value for a physical quantity. It involves the use of instruments, such as thermometers, calipers, and micrometers, to collect data about the physical world.
2. Mechatronics is the combination of mechanical, electrical, and computer engineering to create automated systems. It is used in a variety of applications such as robotics, automation, and control systems.
3. The evolution of measurement and mechatronics has been driven by the need for more accurate and reliable measurements. This has led to the development of more sophisticated instruments and systems, such as digital calipers and automated control systems.
4. Advances in technology have enabled the development of more accurate and reliable measurement systems. For example, the development of digital calipers has allowed for more precise measurements than ever before.
5. Automation has also been a driving force behind the evolution of measurement and mechatronics. Automation allows for the automation of repetitive tasks, such as measuring and controlling a process, and can improve accuracy and reduce costs.
6. The development of computer-aided design (CAD) software has enabled the creation of more complex and accurate designs. CAD software can be used to create detailed models of components and systems, which can then be used to create more efficient and accurate control systems.
7. The development of sensors has enabled the development of more reliable measurement systems. Sensors can be used to measure a variety of physical quantities, such as temperature, pressure, and acceleration.
8. The development of microprocessors has enabled the development of more

sophisticated control systems. Microprocessors can be used to process data from sensors and to control a variety of automated systems.

9. The development of wireless communication has enabled the development of remote monitoring and control systems. This allows for the remote monitoring and control of a variety of automated systems, such as robots and automated manufacturing systems.

Scope for the notes of the Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEER

1. Introduction to Mechatronics: An overview of the principles, components, and applications of mechatronics, with a focus on the integration of mechanical, electrical, and computer engineering.
2. Mechatronics System Design: A look at the design of mechatronic systems, including the selection of components and integration of systems.
3. Measurement and Control: An introduction to the principles and techniques used in the measurement and control of mechatronic systems.
4. Sensors and Actuators: A look at the types of sensors and actuators used in mechatronic systems, as well as the principles and techniques of their operation.
5. Digital Control Systems: An overview of the principles and techniques of digital control systems, and how they are used in mechatronic systems.
6. Robotics: An introduction to the principles and techniques used in robotic systems, with a focus on their use in mechatronic systems.
7. Automation: An overview of the principles and techniques used in automated systems, and how they are used in mechatronic systems.